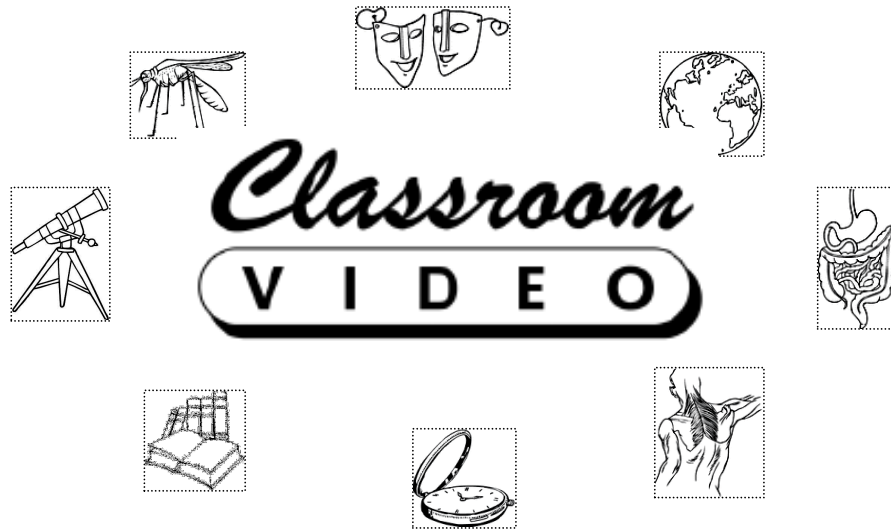




Sustainability in Australia Series
The House With No Bills
A Case Study on Renewable Energy

2008
19 minutes

Program Synopsis:

Some years ago retired engineer Manfred Pruter and his wife Nita moved from Melbourne to a rural property in Central Victoria. By using solar, wind and numerous water tanks the couple have created their own alternative energy paradise. The Pruters supply 100% of their own electricity, heating and water. They now live independently of the energy and water authorities. This inspiring program is a must for anyone interested in renewable energy and sustainability. Learn about the principles of solar, wind and thermal heating and the potential they have to shift society away from fossil fuels.

Teacher Notes:

Rod Williams
B.Ed. (Science)

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Introduction

Manfred and Nita Pruter have shown that it is possible to live a normal life using 100% renewable and sustainable energy. Through a combination of passion, commitment and determination to reduce reliance on fossil fuels, they have set up a household on a rural property in Victoria, Australia, that is self-sufficient in water, electricity, hot water and space heating. 'Windsholme', the Pruter's farm, is an alternative energy paradise, and is indeed, 'the house with no bills'.

Program Worksheet

Before the Program

Find out the meaning of these terms:

- Sustainable energy
- Renewable energy
- Photovoltaic cell
- Wind turbine
- Hebel block
- Double-glazed window

During the Program

1. How much electricity, heating and water does the Pruter house provide for itself?

2. The narrator mentions two effects of climate change on rural communities. What are they?

3. Manfred has devised a unique method for catching extra rainwater. Explain what it is.

4. One millimetre of rain on one square metre of roof yields _____ of water.

5. The average amount spent on energy bills by Australian households, per year, is \$_____.

6. Complete this table outlining the Pruter's household bills.

Bills they do have	Bills they DO NOT have

7. List three static sustainable energy features that have been included in the construction of the Pruter's house.

8. List the five key elements involved in the space heating system.

9. The table below summarises three features of hot water systems.

Most common water heating system in Australia?	
Almost half of the average household energy bill is spent on?	
Amount of greenhouse gases produced by the average water heating system?	

10. Complete the sentences below.

Benefits of a Solar Hot Water System:

- a) Water is heated without emitting _____.
- b) The sun is the source of heat; it's free, giving an _____ benefit.
- c) It requires _____ maintenance.
11. What is the maximum amount of electrical energy produced per day by the Pruter's solar arrays and wind turbines?

12. The word 'photovoltaic' literally means _____

13. Explain the purpose of the tracking system on the solar PV array?

14. These next three questions are about the solar electricity system:

- a) What do the solar PV panels do when sunlight shines on them?

- b) What do the batteries do?

- c) What does the inverter do?

15. Complete the sentences below.

Benefits of a Solar Electricity System:

- a) Electricity is generated without emitting _____.
- b) Electricity is generated from a _____ source – the sun.
- c) Solar PV panels should last _____ years.
- d) It requires _____ maintenance.
- e) When connected to the power grid, excess power produced can gain a _____ on the electricity bill.

16. Write a short paragraph to explain how a wind turbine produces electricity, using the words 'mechanical', 'kinetic' and 'electrical'.

17. List four benefits of a Wind Turbine System

After the Program

1. Investigate grid-connect solar PV systems – find out how they work.
2. Energy companies pay a tariff for electricity produced by grid-connect solar PV systems. Research and compare the grid-connect tariff paid in Australia with the tariff paid in Germany and California.
3. Discuss the advantages and disadvantages of renewable energy systems.
4. Renewable energy systems can be expensive. But for Manfred 'it is just a question of priorities'. What should be the priority for individuals and for governments? Discuss.
5. For further research possibilities, go to <http://www.environment.gov.au/settlements/renewable/index.html>

Suggested Student Responses

During the Program

- How much electricity, heating and water does the Pruter house provide for itself?
100%
- The narrator mentions two effects of climate change on rural communities. What are they?
Dams dry up; stringent water restrictions
- Manfred has devised a unique method for catching extra rainwater. Explain what it is.
Rainfall is collected from a large flat rock platform, filtered and stored in a tank.
- One millimetre of rain on one square metre of roof yields **one litre** of water.
- The average amount spent on energy bills by Australian households, per year, is **\$1500**.
- Complete this table outlining the Pruter's household bills.

Bills they do have	Bills they DO NOT have
Rates	Electricity
Telephone	Hot water
	Heating
	Water

- List three static sustainable energy features that have been included in the construction of the Pruter's house.
Hebel block walls
Double-glazed windows
Thick insulation under the roof
- List the five key elements involved in the space heating system.
Solar hot water panels on roof
Insulated hot water storage tanks under the ground
Underfloor hot water pipes
Electric water pumps and thermostat
Solar photovoltaic panels and batteries
- The table below summarises three features of hot water systems.

Most common water heating system in Australia?	Electric hot water
Almost half of the average household energy bill is spent on?	Heating water
Amount of greenhouse gases produced by the average water heating system?	Four tonnes

- Complete the sentences below.
Benefits of a Solar Hot Water System:
 - Water is heated without emitting **greenhouse gases**.
 - The sun is the source of heat; it's free, giving an **economic** benefit.
 - It requires **low** maintenance.
- What is the maximum amount of electrical energy produced per day by the Pruter's solar arrays and wind turbines?
30 kilowatts

12. The word 'photovoltaic' literally means
light energy
13. Explain the purpose of the tracking system on the solar PV array?
It turns the array around during the day so that the panels are always facing the sun, giving maximum efficiency.
14. These next three questions are about the solar electricity system:
 - a) What do the solar PV panels do when sunlight shines on them?
They produce DC (direct current) electricity.
 - b) What do the batteries do?
Store electricity
 - c) What does the inverter do?
Converts DC (direct current) to AC (alternating current)
15. Complete the sentences below.
Benefits of a Solar Electricity System:
 - a) Electricity is generated without emitting **greenhouse gases**.
 - b) Electricity is generated from a **free** source – the sun.
 - c) Solar PV panels should last **twenty to thirty** years.
 - d) It requires **little** maintenance.
 - e) When connected to the power grid, excess power produced can gain a **credit** on the electricity bill.
16. Write a short paragraph to explain how a wind turbine produces electricity, using the words 'mechanical', 'kinetic' and 'electrical'.
The kinetic energy of the wind spins the vanes - this makes the turbine spin. This mechanical energy then produces electrical energy.
17. List four benefits of a Wind Turbine System.
 - a) **Electricity is generated without emitting greenhouse gases.**
 - b) **Electricity is generated from a free source – the wind.**
 - c) **Wind is an ongoing renewable energy source.**
 - d) **Wind energy can supply power in remote areas.**